

**BALAGANSK, Russian Federation**WMO: **306120**Lat: **54.000N**Lon: **103.067E**Elev: **1400**StdP: **13.97**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -37.1      | -32.0 | -42.2                         | 0.5 | -36.6 | -37.1 | 0.7 | -31.9 | 16.8                  | 5.5  | 13.4 | 8.8  | 2.9                   | 280  | 0.633 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.3                   | 83.2            | 65.3 | 80.1 | 64.4 | 76.9 | 63.5 | 68.7                | 77.9 | 66.9 | 75.5 | 65.2 | 73.7 | 5.7                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.6                            | 99.5 | 73.0 | 63.8 | 93.5 | 71.0 | 62.1 | 87.7 | 69.4 | 33.9          | 77.8 | 32.4 | 75.7 | 31.0 | 73.8 | 75.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>16.4</b>   | <b>13.9</b> | <b>11.9</b> | <b>-38.5</b>               | <b>89.0</b> | <b>7.4</b>         | <b>2.7</b> | <b>-43.8</b>                                       | <b>91.0</b> | <b>-48.2</b> | <b>92.6</b> | <b>-52.4</b> | <b>94.1</b> | <b>-57.8</b> | <b>96.1</b> |    |    |
| (5)               |             |             | <b>-38.2</b>               | <b>71.5</b> | <b>7.0</b>         | <b>2.0</b> | <b>-43.3</b>                                       | <b>73.0</b> | <b>-47.4</b> | <b>74.2</b> | <b>-51.3</b> | <b>75.3</b> | <b>-56.4</b> | <b>76.8</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 30.3   | -9.6  | -4.2  | 13.1  | 34.2  | 47.2  | 60.9  | 66.0  | 61.3  | 48.1  | 33.8  | 13.9  | -3.6  |
|                                                                               | DBStd       | 28.05  | 12.70 | 12.75 | 12.16 | 8.10  | 7.69  | 6.90  | 5.03  | 5.94  | 7.17  | 8.37  | 12.87 | 13.25 |
|                                                                               | HDD50       | 8501   | 1846  | 1516  | 1143  | 476   | 147   | 5     | 0     | 3     | 120   | 502   | 1082  | 1661  |
|                                                                               | HDD65       | 12814  | 2311  | 1936  | 1608  | 924   | 553   | 158   | 48    | 144   | 508   | 966   | 1532  | 2126  |
|                                                                               | CDD50       | 1308   | 0     | 0     | 0     | 2     | 60    | 332   | 498   | 352   | 63    | 1     | 0     | 0     |
|                                                                               | CDD65       | 145    | 0     | 0     | 0     | 0     | 1     | 35    | 80    | 28    | 1     | 0     | 0     | 0     |
|                                                                               | CDH74       | 1369   | 0     | 0     | 0     | 0     | 33    | 425   | 655   | 246   | 10    | 0     | 0     | 0     |
|                                                                               | CDH80       | 278    | 0     | 0     | 0     | 0     | 4     | 94    | 141   | 37    | 2     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 5.2    | 4.0   | 4.1   | 4.9   | 6.2   | 6.4   | 5.7   | 5.2   | 5.2   | 5.6   | 5.5   | 5.2   | 4.3   |
| Precipitation                                                                 | PrecAvg     | 13.3   | 0.4   | 0.3   | 0.2   | 0.6   | 1.0   | 2.0   | 2.6   | 2.7   | 1.8   | 0.7   | 0.5   | 0.6   |
|                                                                               | PrecMax     | 20.3   | 0.7   | 0.6   | 0.6   | 1.2   | 2.3   | 3.9   | 4.6   | 7.1   | 4.7   | 1.4   | 1.0   | 1.5   |
|                                                                               | PrecMin     | 7.5    | 0.1   | 0.0   | 0.0   | 0.2   | 0.2   | 0.4   | 0.7   | 0.7   | 0.4   | 0.1   | 0.2   | 0.1   |
|                                                                               | PrecStd     | 3.1    | 0.2   | 0.2   | 0.2   | 0.3   | 0.5   | 1.0   | 0.9   | 1.5   | 0.9   | 0.3   | 0.2   | 0.3   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.2  | 29.0  | 40.9  | 62.5  | 78.9  | 87.4  | 87.8  | 83.9  | 74.3  | 57.9  | 40.7  | 27.4  |
|                                                                               |             | MCWB   | 22.7  | 26.7  | 34.0  | 47.5  | 57.9  | 64.4  | 66.5  | 65.5  | 60.6  | 49.7  | 36.5  | 25.1  |
|                                                                               | 2%          | DB     | 18.4  | 22.0  | 36.8  | 56.4  | 72.3  | 82.6  | 83.8  | 79.9  | 68.1  | 52.8  | 36.6  | 23.1  |
|                                                                               |             | MCWB   | 17.2  | 19.9  | 31.8  | 43.8  | 54.5  | 63.6  | 66.9  | 65.1  | 57.7  | 46.1  | 33.2  | 21.5  |
|                                                                               | 5%          | DB     | 13.2  | 17.6  | 33.9  | 51.7  | 67.2  | 79.1  | 80.7  | 76.3  | 64.1  | 49.0  | 34.3  | 19.2  |
|                                                                               |             | MCWB   | 12.2  | 16.2  | 29.9  | 41.1  | 51.7  | 62.2  | 66.0  | 63.7  | 55.4  | 43.4  | 31.5  | 18.1  |
|                                                                               | 10%         | DB     | 8.7   | 13.4  | 30.9  | 46.9  | 62.2  | 75.2  | 77.5  | 73.3  | 60.2  | 45.6  | 31.6  | 14.8  |
|                                                                               |             | MCWB   | 7.8   | 12.2  | 27.6  | 38.7  | 49.0  | 60.7  | 65.0  | 62.7  | 53.1  | 40.7  | 29.2  | 13.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.1  | 27.0  | 35.3  | 48.2  | 59.1  | 68.8  | 71.6  | 69.9  | 62.1  | 50.5  | 37.5  | 25.6  |
|                                                                               |             | MCDB   | 23.9  | 29.2  | 39.1  | 59.7  | 76.1  | 79.3  | 80.7  | 77.8  | 71.0  | 57.0  | 40.2  | 27.3  |
|                                                                               | 2%          | WB     | 17.2  | 20.0  | 32.4  | 44.9  | 55.8  | 66.2  | 69.6  | 66.8  | 58.9  | 46.8  | 33.9  | 21.6  |
|                                                                               |             | MCDB   | 18.3  | 21.9  | 36.0  | 54.8  | 69.4  | 76.8  | 79.2  | 75.5  | 67.3  | 51.7  | 36.0  | 22.9  |
|                                                                               | 5%          | WB     | 12.2  | 16.1  | 30.2  | 41.7  | 52.9  | 64.3  | 68.0  | 65.1  | 56.3  | 43.8  | 31.7  | 18.1  |
|                                                                               |             | MCDB   | 13.2  | 17.5  | 33.8  | 50.6  | 65.0  | 75.3  | 76.8  | 74.0  | 62.8  | 48.3  | 34.2  | 19.3  |
|                                                                               | 10%         | WB     | 7.9   | 12.2  | 27.6  | 39.0  | 50.1  | 62.4  | 66.5  | 63.5  | 53.7  | 41.1  | 29.2  | 13.9  |
|                                                                               |             | MCDB   | 8.7   | 13.5  | 30.9  | 46.9  | 60.7  | 72.1  | 74.4  | 71.9  | 59.3  | 45.0  | 31.6  | 14.8  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.5  | 16.2  | 19.0  | 18.2  | 21.6  | 20.7  | 18.3  | 17.8  | 16.8  | 14.4  | 13.0  | 13.3  |
|                                                                               |             | MCDBR  | 15.4  | 18.9  | 18.2  | 25.9  | 30.0  | 27.7  | 24.6  | 24.1  | 22.8  | 19.1  | 12.0  | 15.5  |
|                                                                               |             | MCWBR  | 14.3  | 17.3  | 14.2  | 15.2  | 15.3  | 12.1  | 10.5  | 11.5  | 13.1  | 13.4  | 9.8   | 14.4  |
|                                                                               | 5% WB       | MCDBR  | 15.0  | 18.5  | 17.2  | 23.5  | 27.2  | 22.6  | 19.8  | 21.1  | 21.3  | 18.2  | 11.3  | 15.3  |
|                                                                               |             | MCWBR  | 14.0  | 17.0  | 13.8  | 14.6  | 14.8  | 10.8  | 9.9   | 11.5  | 13.1  | 13.2  | 9.5   | 14.3  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.282  | 0.308 | 0.337 | 0.379 | 0.426 | 0.429 | 0.428 | 0.391 | 0.345 | 0.326 | 0.301 | 0.280 |       |
|                                                                               | taud        | 2.030  | 2.020 | 2.044 | 2.195 | 2.121 | 2.221 | 2.259 | 2.346 | 2.431 | 2.351 | 2.139 | 2.007 |       |
|                                                                               | Ebn at Noon | 202    | 239   | 261   | 268   | 260   | 261   | 259   | 262   | 262   | 238   | 195   | 172   |       |
|                                                                               | Edh at Noon | 23     | 33    | 41    | 40    | 46    | 42    | 40    | 35    | 28    | 24    | 21    | 19    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 344    | 692   | 1204  | 1559  | 1732  | 1908  | 1726  | 1464  | 1022  | 615   | 328   | 236   |       |
|                                                                               | RadStd      | 20     | 32    | 48    | 108   | 142   | 117   | 156   | 84    | 89    | 48    | 28    | 23    |       |